Zero-shot prioritization and mechanistic annotation of MAPT/Tau missense variants

MAPT Zero-Shot Project

Draft reading version

Abstract

MAPT/Tau missense variants are difficult to interpret because many possible substitutions lack clinical or experimental labels. We generated a complete zero-shot atlas for the 441-aa Tau-F / hTau40 isoform by enumerating all 8379 single amino-acid substitutions and scoring them with an ESM-1v ensemble. We added Tau-specific mechanistic annotations and integrated ClinVar and AlphaMissense using strict reference-residue coordinate quality control. The atlas highlights higher zero-shot sensitivity in Tau microtubule repeat regions and prioritizes ClinVar variants of uncertain significance for follow-up. Because the strict ClinVar binary benchmark is small, this work should be interpreted as a reproducible prioritization atlas, not as a standalone clinical diagnostic model.

Keywords: MAPT; Tau; missense variants; zero-shot prediction; protein language models; ESM-1v; AlphaMissense; ClinVar.

1 Introduction

Protein-coding missense variants are common in human genetic data, but many are difficult to interpret. A missense variant changes one amino acid in a protein. Some such changes are harmless, while others disrupt protein function, alter protein interactions, or increase disease risk. For many disease-associated genes, the number of possible missense variants is much larger than the number of variants that have been experimentally or clinically characterized.

MAPT encodes the microtubule-associated protein Tau, a neuronal protein involved in microtubule binding and stabilization. Tau is also central to tauopathies, where abnormal Tau pathology is a defining feature. In inherited frontotemporal dementia and parkinsonism linked to chromosome 17, missense and splice-site mutations in MAPT provided direct genetic evidence that Tau dysfunction can cause neurodegeneration [Hutton et al., 1998, Spillantini et al., 2000].

The biological interpretation of MAPT variation is complicated by Tau isoforms. Adult human brain expresses multiple Tau isoforms generated by alternative splicing. These isoforms differ in N-terminal inserts and in the number of microtubule-binding repeats. The 441-amino-acid 2N4R Tau-F / hTau40 isoform is a commonly used reference for Tau biology, but external databases may report MAPT variants using different transcript or protein coordinate systems. This makes strict coordinate checking essential before combining clinical labels, external predictor scores, and protein-model outputs [Spillantini et al., 2000].

ClinVar provides clinical interpretations of human variants and is useful for benchmarking computational predictions [Landrum et al., 2018]. However, ClinVar is not a complete truth set. Many variants remain variants of uncertain significance, and submitted interpretations can be sparse, conflicting, or dependent on the reference sequence used. For MAPT/Tau-F, strict reference-residue matching leaves only a small binary benchmark.

Protein language models provide a complementary approach. These models are trained on large collections of natural protein sequences and learn statistical patterns related to evolutionary and biochemical constraints [Rives et al., 2021]. ESM-1v showed that pretrained protein language models can be used in a zero-shot setting to predict mutation effects without task-specific experimental training data [Meier et al., 2021]. General missense predictors such as AlphaMissense provide another useful point of comparison, but they also require coordinate quality control in isoform-rich genes such as MAPT [Cheng et al., 2023].

Here, we present a reproducible zero-shot missense atlas for 441-aa Tau-F. We enumerated all 8379 possible single amino-acid substitutions, scored them with an ESM-1v ensemble, added Tau-specific mechanistic annotations, and integrated ClinVar and AlphaMissense through strict reference-residue quality control.

2 Methods

2.1 Study design

This study builds a label-free missense variant atlas for MAPT/Tau. We first listed every possible single amino-acid change in Tau-F, then used pretrained protein language models to score those changes without using clinical labels for training or tuning. Clinical databases were added only after scoring, as evaluation and interpretation layers.

2.2 Tau reference sequence and variant enumeration

All analyses used the 441-amino-acid adult CNS Tau-F / hTau40 / 2N4R isoform as the reference sequence. For each of the 441 positions, all 19 possible non-reference amino-acid substitutions were generated. This produced 8379 missense variants. Each variant was assigned a compact identifier such as P301L, meaning the reference amino acid proline at position 301 is changed to leucine.

2.3 Tau-specific annotation

Each variant was annotated with interpretable Tau biology features, including Tau region, aggregation motif proximity, post-translational modification proximity, known pathogenic hotspot proximity, charge-changing substitutions, and changes involving special residues such as proline, glycine, or cysteine. These annotations were not used to train ESM. They were used for interpretation, prioritization, and a transparent heuristic baseline.

2.4 Zero-shot ESM scoring

Variants were scored with pretrained ESM protein language models using masked marginal scoring. For each Tau-F position, the model estimated the log probability of the reference amino acid and the mutant amino acid in the same sequence context. The raw log-likelihood ratio was

$$\text{esm\_llr} = \log p(\text{mutant}) - \log p(\text{wildtype}).$$

For easier prioritization, we used

$$\text{pathogenic\_score} = -\text{esm\_llr}.$$

Higher scores therefore indicate stronger zero-shot sequence incompatibility. The manuscript-scale run used five ESM-1v checkpoints and averaged their scores.

2.5 ClinVar and AlphaMissense quality control

ClinVar and AlphaMissense annotations were imported after zero-shot scoring. Rows were accepted only if the reported reference amino acid matched the 441-aa Tau-F reference sequence at the mapped position. Rows outside the 1–441 range or with reference-residue mismatches were rejected and written to quality-control tables. This strict matching prevents accidental mixing of MAPT isoform coordinate systems.

2.6 Heuristic baseline and concordance analysis

A transparent Tau heuristic baseline was created to test whether ESM scores only reflected obvious Tau-domain features. We then compared ESM-1v, the heuristic, and AlphaMissense using a top-decile rule: a variant was considered high-priority for a method if it fell in that method's top 10 percent of scored variants.

3 Results

3.1 A complete Tau-F missense atlas

We generated a complete single amino-acid substitution atlas for the 441-residue 2N4R Tau-F / hTau40 reference sequence. The atlas contains 8379 missense variants, corresponding to 19 possible substitutions at each reference position. Clinical annotations were added only after the full variant list and zero-shot model scores had been generated.

3.2 Strict clinical-label coordinate QC limits direct benchmarking

ClinVar import retained 33 annotated variants and rejected 1119 MAPT rows that did not cleanly map to the Tau-F atlas. Accepted labels were sparse: one pathogenic or likely pathogenic variant, two benign or likely benign variants, 26 variants of uncertain significance, and four variants with conflicting classifications. Therefore, the direct P/LP versus B/LB benchmark contains only three examples. AUROC and AUPRC values from this benchmark should be interpreted as pipeline checks rather than clinical performance estimates.

3.3 ESM-1v ensemble scores highlight Tau repeat regions

At the domain level, the highest average ESM-1v ensemble scores were observed in the microtubule repeat regions, while the N-terminal projection domain had the lowest average score (Table 1).

Table 1: Domain-level ESM-1v ensemble summary.

Region	Mean score	Top 10% fraction
microtubule repeat R4	13.22	0.262
microtubule repeat R3	13.21	0.255
microtubule repeat R2 / exon 10	12.96	0.248
microtubule repeat R1	12.64	0.211
C-terminal tail	12.03	0.114
proline-rich region	9.83	0.057
N-terminal projection	3.08	0.000

3.4 Model concordance and discordance

Using the top-decile rule, 209 variants were high-priority for both ESM-1v and the Tau heuristic. These variants were concentrated in Tau repeat regions and included C322E, D314W, S258E, L315D, and I308W. There were also 629 ESM-only high-priority variants and 629 heuristic-only high-priority variants. No variant was high-priority for all three methods under the top-decile rule, primarily because AlphaMissense coverage of the 441-aa Tau-F atlas was limited after strict coordinate quality control.

3.5 ClinVar VUS prioritization

Because most strictly mapped ClinVar variants were VUS, prioritization is more appropriate than binary classification for the current clinical layer. The top ESM-1v-ranked ClinVar VUS candidates included G440E, G333A, K290Q, T377A, and D34Y (Table 2). These variants should be treated as candidates for follow-up review, not as clinical reclassifications.

Table 2: Top ESM-1v-ranked ClinVar VUS candidates.

Rank	Variant	Score	Region
1	G440E	10.37	C-terminal tail
2	G333A	8.73	microtubule repeat R3
3	K290Q	8.48	microtubule repeat R2 / exon 10
4	T377A	7.02	C-terminal tail
5	D34Y	4.12	N-terminal projection

3.6 AlphaMissense coverage

AlphaMissense scores were available for 877 of 8379 Tau-F missense variants, covering 149 positions. Among scored variants, 798 were AlphaMissense likely benign, 67 ambiguous, and 12 likely pathogenic. The importer rejected 3932 MAPT AlphaMissense rows after filtering for P10636: 2048 were outside the 441-aa Tau-F reference range, and 1884 had a reference amino-acid mismatch.

4 Discussion

This study produced a complete zero-shot missense atlas for the 441-aa Tau-F / hTau40 isoform. The atlas combines ESM-1v ensemble scores with Tau-specific mechanistic annotations, ClinVar coordinate QC, AlphaMissense coordinate QC, and ranked VUS prioritization. The strongest claim is not clinical diagnosis. The strongest claim is that the atlas gives researchers a transparent starting point for MAPT/Tau variant review, hypothesis generation, and follow-up prioritization.

The ESM-1v ensemble assigned higher average scores to the microtubule repeat regions than to the N-terminal projection domain. This pattern is biologically plausible because the repeat regions are central to Tau microtubule binding and aggregation-related biology. However, this domain-level signal should be interpreted as a prioritization pattern rather than proof that every high-scoring repeat-region substitution is pathogenic.

A major practical lesson is that coordinate QC is not a minor technical detail. Strict ClinVar and AlphaMissense matching showed that many external MAPT rows cannot be safely mapped to the 441-aa Tau-F reference. This does not mean that ClinVar or AlphaMissense are low-quality resources; it means that MAPT isoform biology makes naive merging risky.

Several limitations remain. First, the clinical benchmark is very small after strict Tau-F coordinate QC. Second, ESM-1v is a sequence model and does not directly model all disease mechanisms, including splicing, isoform expression, post-translational modification state, aggregation kinetics, or patient-level genetic background. Third, AlphaMissense coverage is incomplete under strict Tau-F matching. Fourth, current VUS rankings are hypothesis-generating and have not been experimentally validated.

Future work should refine concordance and discordance analysis into publication-ready supplementary tables and variant-level case studies. Top VUS candidates should be reviewed against population frequency, literature reports, segregation data, functional studies, and domain-specific Tau biology.

5 Figures

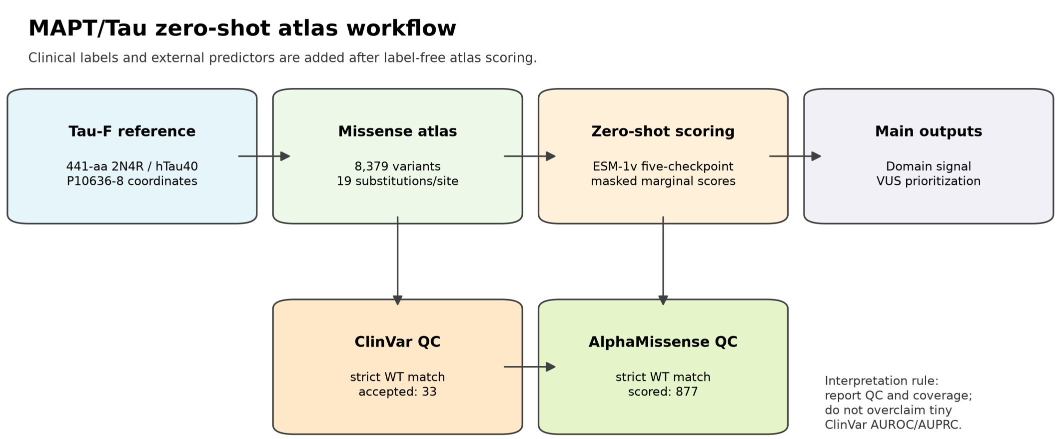

Figure 1: MAPT/Tau zero-shot atlas workflow. Clinical labels and external predictors are added after label-free atlas scoring, with strict coordinate QC.

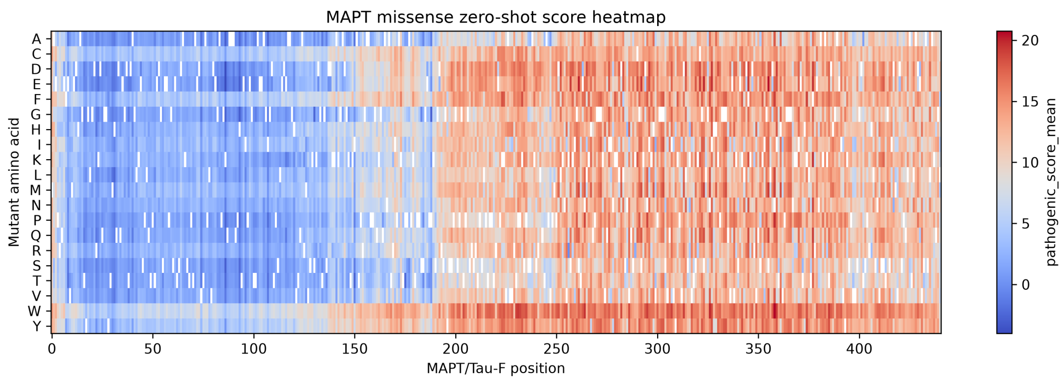

Figure 2: ESM-1v missense score heatmap across Tau-F. Each position corresponds to one amino-acid site and each row to a mutant amino acid.

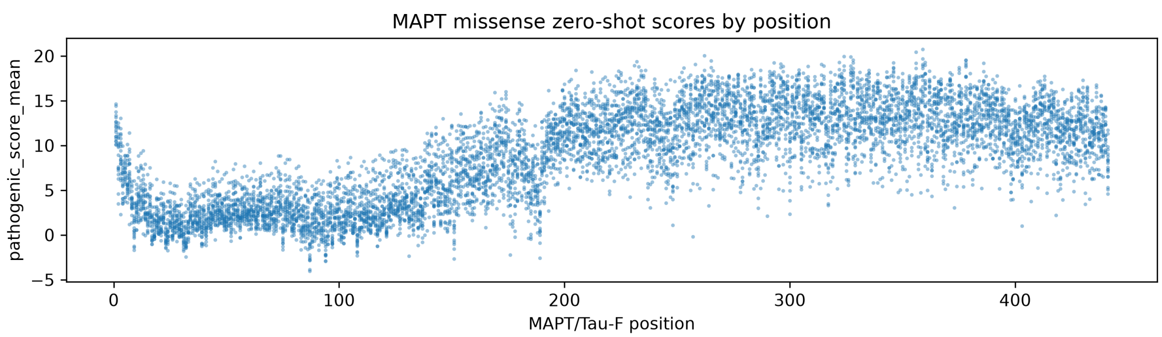

Figure 3: ESM-1v ensemble scores by Tau-F position.

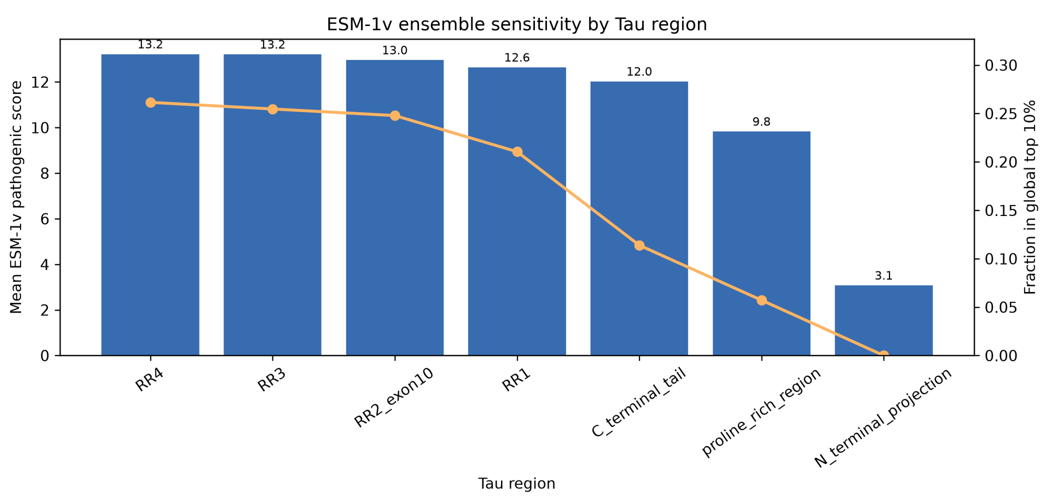

Figure 4: Domain-level ESM-1v score patterns across Tau-F regions.

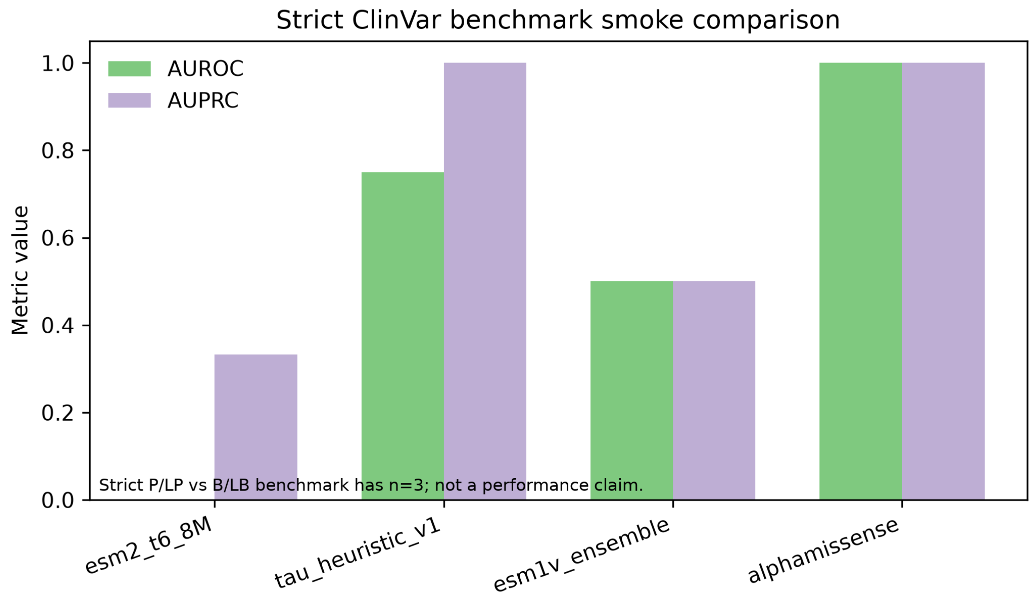

Figure 5: Strict ClinVar smoke benchmark comparison. The benchmark is too small for clinical performance claims.

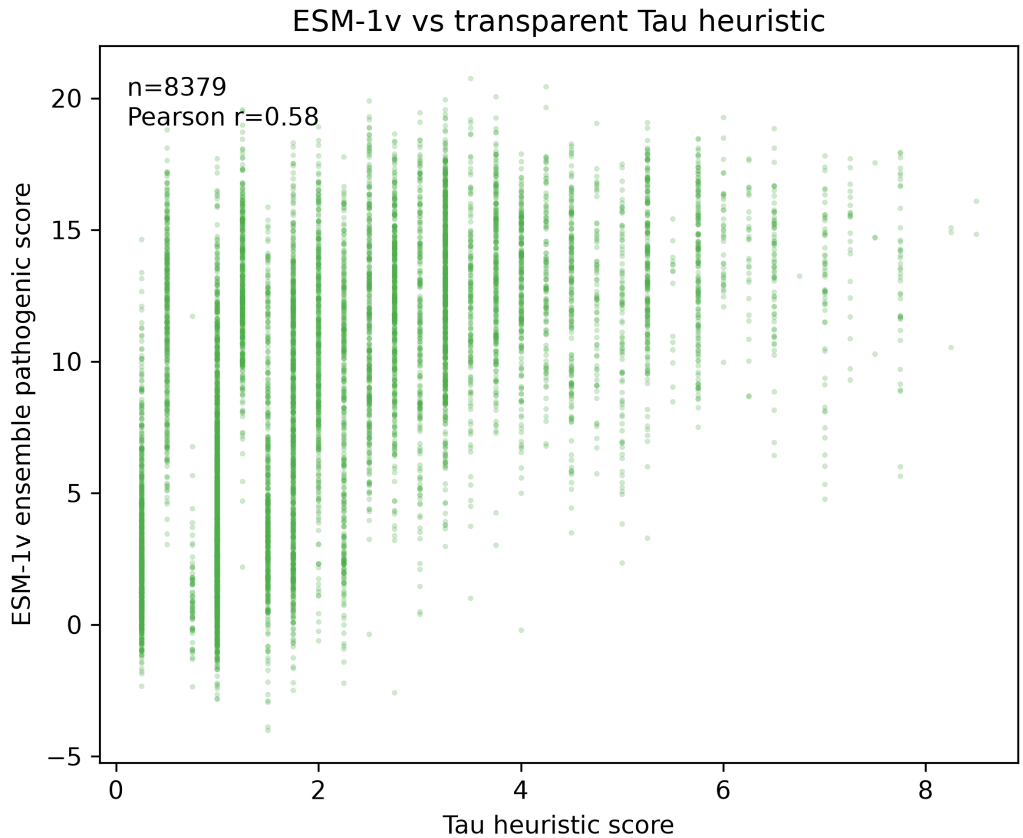

Figure 6: ESM-1v ensemble scores compared with the transparent Tau heuristic baseline across all 8379 variants.

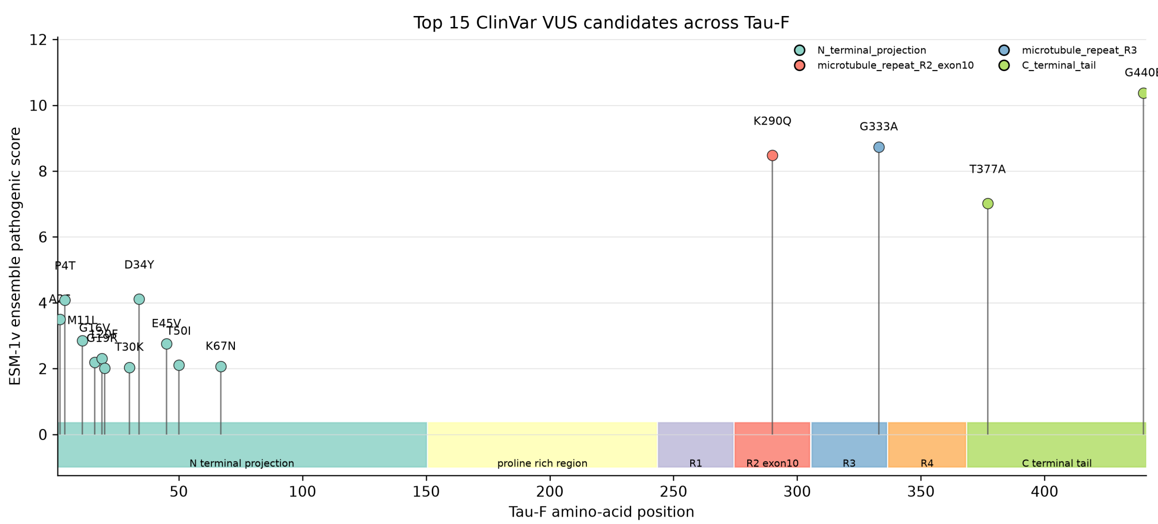

Figure 7: Top ClinVar VUS candidates across Tau-F, ranked by ESM-1v ensemble pathogenic score.

6 Data and code availability

All source code for the MAPT/Tau zero-shot atlas workflow is available at https://github.com/xiaopangF/Tau-zero-shot-MAPT. Large generated files are not committed directly to GitHub. They can be regenerated from public ClinVar, AlphaMissense, and ESM resources using the repository workflow. Before journal submission, generated result tables and figures should be archived with a permanent DOI through a repository such as Zenodo or Figshare.

7 Supplementary tables

Supplementary tables include the full Tau-F missense atlas, ESM-1v ensemble scores, ClinVar coordinate QC tables, AlphaMissense coordinate QC tables, the top VUS priority list, and model concordance/discordance tables.

References

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